

Nguyen Ngoc Hieu

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PROFESSIONAL SUMMARY

Data Science undergraduate experienced in building end-to-end ML and analytics pipelines, from data cleaning and feature engineering to modeling and deployment. Delivered a production-style RAG system, customer-review sentiment/recommendation models, and an HR analytics report with Power BI dashboards. Nationally recognized in data science and competitive programming, with a strong foundation in statistics and algorithms.

TECHNICAL SKILLS

Programming Python, C/C++ , Java , MATLAB
Data Science Pandas, NumPy, Scikit-learn, Machine Learning, Statistical Analysis, Feature Engineering
AI & NLP LLMs, RAG, NLP, Sentiment Analysis, LangChain, ChromaDB
Visualization Power BI, Excel, Matplotlib
Tools Git, GitHub, Jupyter Notebook, Google Colab, LaTeX
Languages Vietnamese (Native), English (Working Proficiency), French A2 (TCF)

EDUCATION

University of Science and Technology of Hanoi (USTH) Jul. 2023 – Sep. 2027 (Expected)
Bachelor of Data Science Hanoi, Vietnam
Relevant Coursework: Machine Learning, Data Mining, Algorithms, Probability & Statistics, Linear Algebra, Database Systems.

PROJECTS

USTH Chatbot – Retrieval-Augmented Generation Nov. 2025 – Feb. 2026
Supervisor: Nguyen Tuan Khang, Regional Leader – IBM AI & Data Platform | GitHub
– Built a local RAG academic assistant for USTH students by converting 9 university PDFs into 185K+ characters and 308 retrievable chunks for document-grounded QA.
– Implemented the retrieval stack with LangChain, ChromaDB, multilingual-e5-base embeddings, OCR support, article-level splitting, and persistent vector storage.

Amazon Electronics Customer Feedback Apr. 2025 – Jun. 2025
Customer Review Analytics and Recommendation Modeling | GitHub
– Cleaned and modeled 1,465 Amazon Electronics product-review records, engineering pricing, discount, rating, and review-text features for customer feedback analysis.
– Built ML/NLP pipelines for satisfaction prediction and sentiment analysis using Random Forest, XGBoost, TF-IDF + SVM, and BERT/DistilBERT-based modeling.
– Implemented product clustering and content-based recommendation with K-Means/DBSCAN and cosine similarity, surfaced through an interactive React analytics dashboard.

HR Performance Analysis May 2025 – Jul. 2025
HR Data Cleaning, Analysis, and Evaluation Project | Project Folder
– Cleaned and analyzed 945 employee-year records for 292 employees across 2014–2019, covering 5 departments, 3 branches, salary, bonus, satisfaction, and performance data.
– Built Excel/Power BI analysis outputs showing 75.7% active employees, 18.2% resignations, average satisfaction score of 3.85 out of 5, and 79.1% “Fully Meets” latest performance ratings.
– Compared compensation and retention indicators by department, branch, status, and job title, including an average total compensation of \$80.5K for HR decision-support insights.

HONORS & AWARDS

– **Third Prize, Data Science Talent Competition** – Vietnam Quant Challenge 2026 **Aug. 2026**
– **Consolation Prize, Vietnamese Student Informatics Olympiad** – Specialized Informatics Division **Dec. 2024**
– **Top 5 Non-IT, ICPC Asia Regional Contest** – ICPC Global **Dec. 2023**
– **Champion, USTH Code Championship** – USTH Youth Union **Apr. 2025**
– **Second Prize, Summer Data BootCamp 2025** – FTU Youth Union **May 2025**
– **Third Prize, Mathematical Modeling Competition** **Aug. 2023**
– **Rector’s Certificate for Students with Outstanding Achievement** – USTH **Dec. 2024**

LEADERSHIP & ACTIVITIES

Class Monitor - Data Science Major, USTH Sep. 2025 – Present
Student Leadership
Represented Data Science students, coordinated academic activities, and supported communication between students and faculty.

Member of Math ICT Department, USTH Learning Support Club Nov. 2023 – Dec. 2025
Learning Platform Development and Tutoring
Developed a grading and online learning platform used by over 1500 students. Organized tutorials and supported first-year students in programming, mathematics, and technical subjects.

Member of R&D Department, USTH Coder Club Dec. 2023 – Jan. 2026
Programming Mentorship
Mentored students in Basic Programming and Data Structures & Algorithms classes, supporting technical training and problem-solving practice.